

On a method for constructing APN permutations

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Abstract

Almost perfect non-linear (APN) permutations are vectorial Boolean functions defined on $\mathbb{F}_{2^n}$ with particular importance in cryptography as well as in some areas of discrete mathematics. In 2009 Browning, Dillon, McQuistan and Wolfe constructed an APN permutation in dimension 6 which is the only example known in an even dimension. In spite of their method being seen as very impactful, it appears that so far this method has not been used to construct other APN permutations. In this abstract we present applications of this method to construct new APN permutations of $\mathbb{F}_{2^n}$ for certain odd n that are not EA equivalent to known APN permutations.

Key Words: APN permutation, CCZ equivalence, affine equivalence, Walsh zero, Gold function.

1 Background

Let $\mathbb{F}_{2^n}$ denote the finite field with 2^n elements and let $\mathbb{F}_{2^n}^*$ be the set of its nonzero elements. Let Tr denote the absolute trace from $\mathbb{F}_{2^n}$ to $\mathbb{F}_2$.

We say that a function $f : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ is *almost perfect non-linear (APN)* if for any $a \in \mathbb{F}_{2^n}^*$ and any $b \in \mathbb{F}_{2^n}$ the equation $f(x + a) + f(x) = b$ has at most two solutions $x \in \mathbb{F}_{2^n}$. A function $f : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ is *affine* if $f(x) = g(x) + c$ where $g(x)$ is $\mathbb{F}_2$ -linear function from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^n}$ and $c \in \mathbb{F}_{2^n}$ is a constant. We say that functions $f : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ and $g : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ are *extended affine (EA) equivalent* if there exist affine permutations L_1, L_2 of $\mathbb{F}_{2^n}$ and an affine mapping $L_3 : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ such that $g(x) = L_1(f(L_2(x))) + L_3(x)$. If it is possible to choose $L_3(x) = 0$ we will say that f and g are *affine equivalent*. EA equivalence is a special case of *Carlet-Charpin-Zinoviev (CCZ) equivalence* [10]. CCZ equivalence preserves the property whether the function is APN or not.

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APN functions are of great interest in cryptography as components of S-boxes in block ciphers, where they provide optimal protection against differential cryptanalysis. They also have connections to combinatorial objects such as finite geometries and combinatorial designs [15].

In some designs of block ciphers, such as substitution-permutation networks [17, Section 4.2], it is desirable that the APN function is also a *permutation* of $\mathbb{F}_{2^n}$.

The main objective of this abstract is to study a known method for finding APN permutations, and to construct new APN permutations, not EA equivalent to known APN permutations, in certain odd dimensions.

Our present knowledge of APN permutations of $\mathbb{F}_{2^n}$ heavily depends on whether n is odd or even. We will survey these two cases in the next two subsections.

1.1 APN permutations in odd dimensions

For odd dimensions, firstly there are six known infinite families of monomial APN permutations, see Table 1 of [7]. In particular, one of these monomial APN families are formed by the *Gold functions* $f(x) = x^{2^i+1}$ on $\mathbb{F}_{2^n}$, where $\gcd(i, n) = 1$. Gold functions appear in several places in this abstract. An important open problem is the conjecture by Dobbertin [13] that any monomial APN function is CCZ equivalent to a permutation belonging to one of these known families.

Overall, all currently known APN permutations in odd dimensions, up to CCZ equivalence, belong to a monomial family mentioned above, or to one of the following families:

- (1) An infinite family of quadratic APN permutations in odd dimension found in 2008 by Budaghyan, Carlet, and Leander [6].
- (2) Two quadratic APN permutations in dimension 9 found in 2020 by Beierle and Leander [2].

As such, it is certainly still desirable to find new constructions of APN permutations in odd dimensions as well.

1.2 APN permutations in even dimensions

In comparison to odd dimensions, the situation is very different in even dimensions. It is known that no APN permutations of $\mathbb{F}_{2^n}$ exist when $n = 2, 4$. An APN permutation of $\mathbb{F}_{2^6}$ was announced at the Fq9 conference in 2009 [5]; we give more details on it in Section 2.1 below. It is an important question whether APN permutations exist in even dimensions greater than 6. This is known as the *Big APN Problem*.

An algorithm for checking CCZ equivalence to a permutation has been implemented in the *sboxU* tool [16]. This implementation has already been

used to verify that all of the known quadratic APN functions in dimension 8 are inequivalent to a permutation.

2 Permutations CCZ equivalent to a given function

The construction of APN permutation of $\mathbb{F}_{2^6}$ due to Browning, Dillon, McQuistan and Wolfe in [5] rests on the following main idea. Start with an APN function, say κ , on $\mathbb{F}_{2^n}$ that does not need to be a permutation. By a certain detailed analysis of the Walsh transform values of κ , construct a function, say f , on the same field $\mathbb{F}_{2^n}$ that is CCZ equivalent to κ , and moreover f is a permutation. Since CCZ equivalence preserves the APN property, it follows that f is an APN permutation of $\mathbb{F}_{2^n}$.

In [5] this method is presented mostly by computing an explicit example on $\mathbb{F}_{2^6}$. In Section 3 below we review some results of Chase and Lisoněk who gave in [12] a theoretical description of the method that is not tied to a specific initial function κ .

2.1 Kim type functions

For the APN permutation in dimension 6, the paper [5] starts with the APN function

$$\kappa(x) = x^3 + x^{10} + ux^{24} \quad (1)$$

where u is a suitable primitive element of $\mathbb{F}_{2^6}$, and it applies the method briefly outlined above. Here the function κ is known as *Kim function*. Considerable efforts were spent by several authors to generalize this result from dimension 6 to higher even dimensions. It was quickly realized that the function κ in (1) naturally belongs to a certain family of quadrimonomials, which were subject of these investigations. Ultimately, Chase and Lisoněk in [11] proved the following result.

Theorem 2.1. [11, Theorem 2.1] *Suppose that $m \geq 4$ is an integer and let $q = 2^m$. Let $f : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_{q^2}$ be given by $f(x) = x^{3q} + a_1x^{2q+1} + a_2x^{q+2} + a_3x^3$ where $a_1, a_2, a_3 \in \mathbb{F}_{q^2}$. If f is APN, then f is affine equivalent to $G_1(x) = x^3$ or f is affine equivalent to $G_2(x) = x^{2^{m-1}+1}$.*

Combined with the result of Göloğlu and Langevin [14] which states that, for even n , Gold APN functions are never CCZ equivalent to permutations, it follows that for $m \geq 4$ Kim-type functions are never CCZ equivalent to permutations, hence the specific result in [5] does not generalize to higher even dimensions.

2.2 Dillon's hexanomial functions

Very recently Bartoli, Grimaldi and Stănică published the preprint [1] in which they study a class of hexanomial functions proposed by Dillon in 2006. These functions are defined over $\mathbb{F}_{q^2}$ where $q = 2^n$ and they have the form

$$F(x) = x(Ax^2 + Bx^q + Cx^{2q}) + x^2(Dx^q + Ex^{2q}) + x^{3q}.$$

This class of functions includes Kim type functions from Section 2.1 which result as the special case $B = E = 0$. The authors of [1] report that so far no APN permutations have been found in their work.

3 General description of the construction method

Originally the process of obtaining a CCZ equivalent permutation for a given function was described in [5] mostly in a computational setting for the specific example obtained in dimension 6. Chase and Lisoněk gave a more general and theoretical description of it in [12]. We now review the main steps of the method as given in [12].

Let f be a function from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^n}$. For $(a, b) \in \mathbb{F}_{2^n} \times \mathbb{F}_{2^n}$ we define the Walsh transform of f at (a, b) as $\mathcal{W}_f(a, b) = \sum_{x \in \mathbb{F}_{2^n}} (-1)^{\text{Tr}(ax + bf(x))}$. We say that (a, b) is a *Walsh zero* of f if $\mathcal{W}_f(a, b) = 0$.

WZ spaces defined below occur implicitly in [5]. Their importance and applications were recognized more explicitly, see [8] and references therein.

Definition 3.1. Let f be a function from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^n}$. Suppose that S is an $\mathbb{F}_2$ -linear subspace of $\mathbb{F}_{2^n} \times \mathbb{F}_{2^n}$ such that $\dim_{\mathbb{F}_2} S = n$ and each element of S except $(0, 0)$ is a Walsh zero of f . We say that S is a *WZ space* of f .

We say that two WZ spaces S, T of the same function *intersect trivially* if $S \cap T = \{(0, 0)\}$.

We now give the main construction as the proof given in [12] of the following proposition. We include the construction here since it is the main purpose of this abstract to investigate the results obtained by this method in certain cases.

Proposition 3.2. [12] *Let f be a function from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^n}$. If there exist two WZ spaces of f that intersect trivially, then f is CCZ equivalent to a permutation of $\mathbb{F}_{2^n}$.*

Proof. Without loss of generality we can assume that $f(0) = 0$, since this can be achieved by an affine transformation which is a special kind of CCZ equivalence.

Let $\{\alpha_1, \dots, \alpha_n\}$ and $\{\beta_1, \dots, \beta_n\}$ be two dual bases of $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$, that is, $\text{Tr}(\alpha_i \beta_j) = \delta_{i,j}$ where for $1 \leq i, j \leq n$ we let $\delta_{i,j} = 1$ if $i = j$ and $\delta_{i,j} = 0$ otherwise. Let the nonzero elements of $\mathbb{F}_{2^n}$ be labelled $x_1, \dots, x_{2^n-1}$.

Let $G = \begin{pmatrix} G_1 \\ G_2 \end{pmatrix}$ be a $(2n) \times (2^n - 1)$ matrix over $\mathbb{F}_2$ where G_1 and G_2 are $n \times (2^n - 1)$ matrices over $\mathbb{F}_2$ defined as follows. The entry in row i and column j of G_1 is $\text{Tr}(\alpha_i x_j)$. The entry in row i and column j of G_2 is $\text{Tr}(\alpha_i f(x_j))$. Then the j -th column of G is of the form $\begin{pmatrix} x_j \\ f(x_j) \end{pmatrix}$ where x_j and $f(x_j)$ are represented as n -dimensional column vectors with respect to the basis $\{\beta_1, \dots, \beta_n\}$ introduced above. Let C be the binary linear code which is the row space of G . Each codeword of C is of the form $\text{Tr}(rx_j + sf(x_j))_{j=1, \dots, 2^n-1}$ where r, s are fixed elements of $\mathbb{F}_{2^n}$.

Let S and T be the two given trivially intersecting WZ spaces, and let $B_1 = \{(a_1, b_1), \dots, (a_n, b_n)\}$ and $B_2 = \{(a_{n+1}, b_{n+1}), \dots, (a_{2n}, b_{2n})\}$ be their bases. Note that $B_1 \cup B_2$ is a basis for $\mathbb{F}_{2^n} \times \mathbb{F}_{2^n}$.

Let $G' = \begin{pmatrix} G'_1 \\ G'_2 \end{pmatrix}$ be a $(2n) \times (2^n - 1)$ matrix over $\mathbb{F}_2$ where G'_1 and G'_2 are $n \times (2^n - 1)$ matrices over $\mathbb{F}_2$ defined as follows. The entry in row i and column j of G'_1 is $\text{Tr}(a_i x_j + b_i f(x_j))$. The entry in row i and column j of G'_2 is $\text{Tr}(a_{n+i} x_j + b_{n+i} f(x_j))$. For $i = 1, 2$ let C'_i be the row space of G'_i . Since S and T are WZ spaces for f , each nonzero codeword of C'_i has weight 2^{n-1} for $i = 1, 2$. Thus C'_1, C'_2 are simplex codes S_n , and G'_1, G'_2 are formed by pairwise distinct nonzero columns. Thus, with respect to the basis $\{\beta_1, \dots, \beta_n\}$, the columns of G' can be viewed as $\begin{pmatrix} x \\ g(x) \end{pmatrix}$ where x runs through all nonzero elements of $\mathbb{F}_{2^n}$ and $g(x) \in \mathbb{F}_{2^n}$. After letting $g(0) = 0$ we see that g is a permutation of $\mathbb{F}_{2^n}$.

Let C' be the rowspace of G' . Since $B_1 \cup B_2$ is a basis for $\mathbb{F}_{2^n} \times \mathbb{F}_{2^n}$, each codeword of C' is of the form $\text{Tr}(rx_j + sf(x_j))_{j=1, \dots, 2^n-1}$ where r, s are fixed elements of $\mathbb{F}_{2^n}$. Thus the codes C and C' are equal, and functions f and g are CCZ equivalent by [4], see also paragraph 4 on page 380 in [9]. $\square$

4 Finding new APN permutations

The method of [5] is widely recognized as impactful, and it is therefore surprising that it does not seem to have been applied to obtain more APN permutations; at least we are not aware of any such results. Therefore it is the main purpose of this abstract to investigate the applicability of this method in other situations, in order to find new APN permutations.

We see in the statement of Proposition 3.2 that it requires a pair of trivially intersecting (abbreviated TI henceforth) WZ spaces as an ingredient. In [12] we gave four general constructions of TI pairs of WZ spaces of the Gold functions $f(x) = x^{2^i+1}$, with $\gcd(i, n) = 1$, defined on $\mathbb{F}_{2^n}$.

At the outset of the work reported in this abstract, we performed very extensive numerical studies of APN permutations resulting from these four

constructions of TI pairs of WZ spaces, which are given as Propositions 5.1 through 5.4 in [12]. The aim was to find APN permutations that can be easily demonstrated as EA inequivalent to Gold functions, which have algebraic degree 2, and also EA inequivalent to the compositional inverses of Gold functions, which have algebraic degree $(n+1)/2$ on $\mathbb{F}_{2^n}$. We recall that Gold functions are injective only when n is odd, and quite to the contrary Gold functions are not CCZ equivalent to permutations when n is even due to the previously mentioned result of Göloğlu and Langevin [14].

Propositions 5.1 through 5.4 in [12] are intentionally stated there in the most general way that authors were able to find, in order to construct as many TI pairs as possible. These propositions contain free parameters valued over the domain of the function which is $\mathbb{F}_{2^n}$, or over its subfield $\mathbb{F}_{2^3}$. Therefore exhaustive computations are impossible already for modest values of n , and we used uniform sampling from the parameter spaces instead. All four Propositions 5.1 through 5.4 in [12] were tested. Proposition 5.3 was the only one that produced APN permutations with algebraic degree different from 2 and $(n+1)/2$, and therefore we focus on it henceforth. We first recall its full statement.

Proposition 4.1. *[12, Proposition 5.3] Let $n = 3k$ where k is odd and $f : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$, $f(x) = x^{2^i+1}$, with $\gcd(i, n) = 1$. Let*

$$Y = \{(x, \mu^{-(2^i+1)}(\xi \text{Tr}(\mu x) + \text{Tr}(\xi^{2^i} \mu x))) : x \in \mathbb{F}_{2^n}\}$$

where ξ is a fixed element of $\mathbb{F}_{2^3} \subset \mathbb{F}_{2^n}$ and μ is a fixed element of $\mathbb{F}_{2^n}^$. Let*

$$Z = \{(\nu(b + b^{\frac{1}{2^i}}), \nu^{2^i+1}b) : b \in \mathbb{F}_{2^n}\}$$

with ν a fixed element of $\mathbb{F}_{2^n}^$. Suppose also that $\text{Tr}((\xi + \xi^{2^i})(\mu\nu)^{-2^i}) = 0$. Then Y and Z are WZ spaces of f and the pair $\{Y, Z\}$ intersects trivially.*

Definition 4.2. We denote by $f_{\xi, \mu, \nu}$ the APN permutation that is obtained from Proposition 3.2 and Proposition 4.1 for $i = 1$ and for specific admissible choices of the parameters ξ , μ and ν .

We note that two bases of $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$ also need to be specified in order to determine $f_{\xi, \mu, \nu}$ uniquely, see the proof of Proposition 3.2. However after a change of basis the new function remains in the same EA equivalence class. When we give an explicit example of $f_{\xi, \mu, \nu}$, the particular choices of these two bases will be also given.

We have the following lemma that reduces the number of parameter triples that need to be investigated.

Lemma 4.3. *For each $\xi \in \mathbb{F}_{2^3}$ and each $\mu, \nu \in \mathbb{F}_{2^n}$ we have $f_{\xi, \mu, \nu} = f_{\xi+1, \mu, \nu}$.*

Proof. Let Y_0 and Y_1 be the Y spaces in the proof of Proposition 4.1 that were used to construct $f_{\xi,\mu,\nu}$ and $f_{\xi+1,\mu,\nu}$ respectively. We have

$$\begin{aligned} & \mu^{-(2^i+1)}((\xi+1)\text{Tr}(\mu x) + \text{Tr}((\xi+1)^{2^i}\mu x)) \\ = & \mu^{-(2^i+1)}(\xi\text{Tr}(\mu x) + \text{Tr}(\mu x) + \text{Tr}(\xi^{2^i}\mu x) + \text{Tr}(\mu x)) \\ = & \mu^{-(2^i+1)}(\xi\text{Tr}(\mu x) + \text{Tr}(\xi^{2^i}\mu x)) \end{aligned}$$

and therefore $Y_0 = Y_1$. Consequently $f_{\xi,\mu,\nu} = f_{\xi+1,\mu,\nu}$. $\square$

4.1 Examples of new APN permutations

To further shorten the process of obtaining and analyzing new APN permutations we can fix the values of parameters μ and ν to suitable constants. For example, taking $\mu = 1$ and $\nu = 1$ leads to admissible parameter choices, since for any ξ we then have

$$\text{Tr}((\xi + \xi^{2^i})(\mu\nu)^{-2^i}) = \text{Tr}(\xi + \xi^{2^i}) = 0$$

and the condition of Proposition 4.1 is satisfied for each ξ .

With these particular choices for μ and ν we get the following results.

Proposition 4.4. *For each $\xi \in \mathbb{F}_{2^3} \setminus \mathbb{F}_2$ the function $f_{\xi,1,1}$ defined on $\mathbb{F}_{2^9}$ is an APN permutation of algebraic degree 4.*

Proposition 4.5. *For each $\xi \in \mathbb{F}_{2^3} \setminus \mathbb{F}_2$ the function $f_{\xi,1,1}$ defined on $\mathbb{F}_{2^{15}}$ is an APN permutation of algebraic degree 4.*

Propositions 4.4 and 4.5 are proved by application of Proposition 3.2, with the WZ spaces provided by Proposition 4.1, and interpolating the univariate polynomial forms of the functions $f_{\xi,1,1}$ in the computer algebra system Magma [3]. Dimensions 9 and 15 occurring in Propositions 4.4 and 4.5 are the smallest two dimensions in which Proposition 4.1 is applicable, as it requires the dimension to be of the form $n = 6k + 3$ with k a positive integer.

An explicit example of one of the functions $f_{\xi,1,1}$ on $\mathbb{F}_{2^9}$ is given in Appendix A at the end of this abstract.

5 Work in progress

Work reported in this abstract is presently in progress, with more results to be included in the full paper. In particular, the following objectives are pursued currently:

- Verify computationally that the functions $f_{\xi,1,1}$ have algebraic degree 4 also in higher applicable dimensions, that is, $n = 6k + 3$ for

$k > 2$, and thus they are not EA equivalent to Gold functions and their inverses in those dimensions either. While a full univariate polynomial interpolation of $f_{\xi,1,1}$ from all 2^n preimages would cause prohibitively extensive computations for larger n , we will take advantage of our algebraic degree conjecture and only interpolate terms of algebraic degree 4 or less, of which there are only $n^4/24 + O(n^3)$, a quantity that remains manageable for larger n . Thus, we will interpolate a *sparse* univariate polynomial. Then we will verify that the interpolated polynomial gives the correct values of $f_{\xi,1,1}$ also on many other preimages that were not used in the interpolation, and thus verify probabilistically that the sparse polynomial in fact gives the function $f_{\xi,1,1}$.

- Examine the influence of the choice of the bases of $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$ (see the proof of Proposition 3.2) on the resulting polynomial form of $f_{\xi,1,1}$. While all such polynomial forms will be EA equivalent, they still may differ substantially in their complexity and shape, and some of them may be more amenable to a theoretical mathematical computer-free analysis. Computer-free proofs of the algebraic degree and of the APN permutation property would be the ultimate goals, as they may then inspire other computer-free constructions.
- Investigate all other APN permutations constructed using the method presented above. In cases where these functions have algebraic degree 2 or $(n+1)/2$, determine whether or not they are EA equivalent to Gold functions or their inverses.

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A Example of APN permutation in dimension 9

In this appendix we present a numerical example of one of the APN permutations in dimension 9 that are constructed in Proposition 4.4. The example

was computed using computer algebra system Magma [3]. In order to make the example reproducible, in the construction we use the default values defined by Magma. Let u and z be the primitive elements of $\mathbb{F}_{2^9}$ and of $\mathbb{F}_{2^3}$ respectively, as defined in Magma. In the notation of the proof of Proposition 3.2, we use for $\{\beta_1, \dots, \beta_n\}$ the default basis of $\mathbb{F}_{2^9}$ over $\mathbb{F}_2$ defined by Magma, which is the polynomial basis $\{1, u, u^2, \dots, u^8\}$, and for $\{\alpha_1, \dots, \alpha_n\}$ we use its dual basis. In Proposition 4.4 we take $\xi = z$. The APN permutation $f_{z,1,1}$ is displayed on the next page as a univariate polynomial over $\mathbb{F}_{2^9}$.

$$\begin{aligned}
f_{z,1,1}(x) = & u^{372}x^{480} + u^{107}x^{464} + u^{502}x^{456} + u^{429}x^{452} + u^{486}x^{450} + u^{445}x^{449} + u^{177}x^{448} \\
& + u^{140}x^{424} + u^{305}x^{420} + u^{13}x^{418} + u^{252}x^{417} + u^{319}x^{416} + u^{154}x^{408} + u^{13}x^{404} \\
& + u^{232}x^{402} + u^{252}x^{401} + u^{431}x^{400} + u^{209}x^{396} + u^{457}x^{394} + u^{106}x^{393} + u^{151}x^{392} \\
& + u^{259}x^{389} + u^{212}x^{388} + u^{80}x^{387} + u^{243}x^{386} + u^{471}x^{385} + u^{123}x^{384} + u^{399}x^{368} \\
& + u^{487}x^{360} + u^{414}x^{356} + u^{52}x^{354} + u^{317}x^{352} + u^{141}x^{344} + u^{179}x^{337} + u^{456}x^{336} \\
& + u^{254}x^{332} + u^{204}x^{329} + u^{375}x^{328} + u^{201}x^{326} + u^{131}x^{325} + u^{69}x^{324} + u^{340}x^{323} \\
& + u^{97}x^{322} + u^{375}x^{321} + u^{34}x^{320} + u^{142}x^{312} + u^{283}x^{308} + u^{502}x^{306} + u^{361}x^{305} \\
& + u^{319}x^{304} + u^{181}x^{300} + u^{131}x^{297} + u^{413}x^{296} + u^{201}x^{294} + u^{58}x^{293} + u^{22}x^{291} \\
& + u^{254}x^{290} + u^{335}x^{289} + u^5x^{288} + u^{254}x^{284} + u^{364}x^{282} + u^{41}x^{281} + u^{437}x^{280} \\
& + u^{420}x^{278} + u^{140}x^{277} + u^{487}x^{276} + u^{241}x^{275} + u^{203}x^{274} + u^{337}x^{273} + u^{78}x^{272} \\
& + u^{449}x^{270} + u^{483}x^{268} + u^{11}x^{267} + u^{468}x^{266} + u^{408}x^{265} + u^{89}x^{264} + u^{372}x^{263} \\
& + u^{212}x^{262} + u^{252}x^{261} + u^{409}x^{260} + u^{194}x^{259} + u^{45}x^{258} + u^{381}x^{257} + u^{283}x^{256} \\
& + u^{80}x^{240} + u^{380}x^{232} + u^{500}x^{228} + u^{208}x^{226} + u^{325}x^{225} + u^{364}x^{224} + u^{457}x^{212} \\
& + u^{191}x^{210} + u^{45}x^{209} + u^{407}x^{208} + u^{217}x^{204} + u^{45}x^{202} + u^{410}x^{201} + u^{173}x^{200} \\
& + u^{349}x^{198} + u^{76}x^{197} + u^{117}x^{196} + u^{319}x^{194} + u^{194}x^{193} + u^{429}x^{192} + u^{15}x^{184} \\
& + u^{502}x^{180} + u^{210}x^{178} + u^{364}x^{177} + u^{393}x^{176} + u^{201}x^{172} + u^{382}x^{168} + u^{420}x^{166} \\
& + u^{131}x^{164} + u^{128}x^{163} + u^8x^{162} + u^{216}x^{161} + u^{408}x^{160} + u^{473}x^{156} + u^{43}x^{154} \\
& + u^{408}x^{153} + u^{407}x^{152} + u^{128}x^{150} + u^{425}x^{148} + u^{10}x^{147} + u^{420}x^{145} + u^{334}x^{144} \\
& + u^{158}x^{142} + u^{141}x^{141} + u^{391}x^{140} + u^{375}x^{139} + u^{502}x^{138} + u^{154}x^{137} + u^{438}x^{136} \\
& + u^{478}x^{135} + u^{218}x^{134} + u^{15}x^{133} + u^{491}x^{132} + u^{319}x^{131} + u^{136}x^{130} + u^{510}x^{129} \\
& + u^{359}x^{128} + u^{445}x^{120} + u^{426}x^{116} + u^{317}x^{114} + u^{171}x^{113} + u^{282}x^{112} + u^{76}x^{106} \\
& + u^{49}x^{105} + u^{45}x^{104} + u^{194}x^{102} + u^{487}x^{101} + u^{302}x^{100} + u^{227}x^{99} + u^{465}x^{98} \\
& + u^{375}x^{97} + u^{465}x^{96} + u^{499}x^{92} + u^{408}x^{90} + u^{262}x^{89} + u^{429}x^{88} + u^{128}x^{86} \\
& + u^{246}x^{84} + u^{375}x^{83} + u^{13}x^{82} + u^{144}x^{81} + u^{212}x^{80} + u^{12}x^{78} + u^{327}x^{77} \\
& + u^{45}x^{76} + u^{229}x^{75} + u^{479}x^{74} + u^{28}x^{72} + u^{178}x^{71} + u^{179}x^{70} + u^{408}x^{69} \\
& + u^{331}x^{68} + u^{399}x^{67} + u^{150}x^{66} + u^{474}x^{65} + u^{473}x^{64} + u^{372}x^{60} + u^{380}x^{58} \\
& + u^{215}x^{57} + u^{340}x^{56} + u^{52}x^{53} + u^{510}x^{52} + u^{60}x^{51} + u^{27}x^{50} + u^{327}x^{49} + u^{227}x^{48} \\
& + u^{161}x^{46} + u^{254}x^{45} + u^{158}x^{44} + u^{179}x^{43} + u^{407}x^{42} + u^{45}x^{41} + u^{319}x^{40} \\
& + u^{105}x^{39} + u^{510}x^{38} + u^{428}x^{37} + u^{173}x^{36} + u^{506}x^{35} + u^{218}x^{34} + u^{402}x^{33} \\
& + u^{224}x^{32} + u^{80}x^{30} + u^{308}x^{29} + u^{216}x^{28} + u^{34}x^{26} + u^{261}x^{25} + u^{209}x^{24} \\
& + u^{454}x^{23} + u^{265}x^{22} + u^{76}x^{21} + u^{375}x^{20} + u^{34}x^{19} + u^{65}x^{18} + u^{278}x^{17} + u^{392}x^{16} \\
& + u^{445}x^{15} + u^{128}x^{14} + u^{408}x^{13} + u^{331}x^{12} + u^{464}x^{11} + u^{319}x^{10} + u^{219}x^9 + u^{207}x^8 \\
& + u^{374}x^7 + u^{436}x^6 + u^{407}x^5 + u^{472}x^4 + u^{330}x^3 + u^{262}x^2 + u^{322}x
\end{aligned}$$